VITyarthi | Open Source Software

The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5 Max Marks: 100

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1. Introduction

In this report, I am conducting a detailed audit of Firefox, the open-source web browser created by the Mozilla Foundation. This is not just an exploration of its technical aspects; it's also a look at its philosophy as a tool designed for user freedom and privacy.

I will explain why Firefox exists, the problem it was designed to address, the freedoms provided by its Mozilla Public License, the ethical values it upholds, and the extensive ecosystem it has developed over the years. Along with my report, I have created five shell scripts that showcase practical Linux skills. These scripts handle system identification, package inspection, permission tracking, log analysis, and user interaction. They reflect the open-source belief in sharing and automation.

Every tool we use as developers is built on open-source foundations. Understanding the philosophy behind Firefox is not optional; it is crucial for any professional.

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2. Choosing Firefox

I selected Firefox for this project. Firefox is an open source web browser developed by Mozilla Foundation. It is licensed under MPL 2.0. I chose Firefox because it is a browser I use every day, and I wanted to learn more about its origins, license, and the open source philosophy behind it .

3. Project Report

Part A — Origin and Philosophy (Units 1 & 2)

A1. The problem firefox was created to solve

Before Firefox, Internet Explorer ruled the web browser market. Microsoft included Internet Explorer with Windows, and most people had no other good choices. Internet Explorer was slow, had many security issues, and did not properly follow web standards. Developers found it hard to create websites that worked well across different browsers.

In the late 1990s, Netscape Navigator was a popular browser. Netscape chose to open source its code, which led to the Mozilla project. This was a bold move; a company was giving away its browser code for free. The Mozilla project aimed to build a better browser that anyone could use and improve.

The turning point came in 2002. The Mozilla project released Phoenix, which later became Firefox. The goal was simple: build a browser that was fast, secure, and respected user choice. Unlike Internet Explorer, Firefox was open source. Anyone could examine the code, find bugs, and suggest improvements.

Firefox introduced features we now consider standard. Tabbed browsing allowed users to open multiple pages in one window. The search bar made it easy to look things up without visiting a search engine first. Pop-up blocking eliminated annoying ads. These features quickly made Firefox popular.

By 2004, Firefox had millions of users. It proved that open source software could compete with offerings from the biggest tech companies. The Mozilla Foundation, a non-profit organization, still develops Firefox today. Their mission is to keep the internet open and accessible for everyone.

A2. The license — what it actually says

Firefox is distributed under the Mozilla Public License 2.0, or MPL 2.0 for short. This license has been specifically chosen to balance the interests of users, developers, and companies that want to use Firefox’s code for their own projects.

The Four Freedoms of Free Software:-

The Free Software Foundation bases its definition of free software on four fundamental freedoms, and Firefox, under the MPL 2.0 license, respects all four of these freedoms.

Freedom 0 - The freedom to run the program for any purpose

The MPL 2.0 places no restrictions on the use of Firefox whatsoever. A student who is interested in learning how to program for the web, a non-profit organization, or a government agency can all run Firefox without needing to obtain permission or pay a fee.

Freedom 1 - The freedom to study and modify the source code

Firefox's source code is available for everyone to download from our GitHub repository, and we encourage users to modify the source code to suit their needs.

Freedom 2 - The freedom to redistribute copies

You are free to distribute Firefox to anyone you want, whether for free or for a fee. You can copy Firefox to a USB drive and give it to a friend. Universities can host Firefox on their mirrors. Linux distributions can include Firefox in their package repositories.

Freedom 3 - The freedom to distribute modified versions

When someone modifies Firefox - for example, to add a new feature or remove tracking - they are free to share their modified version with the world. This is how projects like Tor Browser and LibreWolf exist. They are built on Firefox code but customized for specific needs.

Commercial Use and copyleft

MPL 2.0 is considered a copyleft license, but it is not as strict as GPL. If the company modifies the files of Firefox and sells the modified version of the program, then the company has to share the modifications with anyone who requests them, but the modifications have to be done under the MPL 2.0 license.

The company, however, can add their own proprietary code with the Firefox code, provided the Firefox code and the proprietary code are kept separately.

For example, the company can develop their own web browser using Firefox and add their own proprietary code to the program and sell the program. They have to share the modifications done to the Firefox code, but the proprietary code can remain closed-source.

GPL v2 vs MPL 2.0 (Key Differences)

Dimension	**GPLv2**	**MPL 2.0**
Copy strength	Strong- all derivatives must be GPL.	File based only modified MPL files must stay MPL
Combining with proprietary code	Restricted, often requires GPL for entire work.	Allowed, as long as MPL files remain separate.
Philosophy	All software should be free.	Balance between freedom and commercial use.

Free as in Free Beer vs Free as in Freedom

One of the most important concepts in open source is understanding free. Firefox is free as in free beer - it costs nothing to download and use. Anyone can go to mozilla.org and get the latest version without paying a cent.

But the real value of Firefox is free as in freedom. If Mozilla disappeared tomorrow, the code would still exist. Anyone could continue developing it. Users are not locked into Mozilla's decisions. This freedom is embedded in both the license and the open development model. No single company controls the future of Firefox.

A3. The ethics of open source

Should All Software Be Open Source?

I don't think that all software should be open source. This is because some pieces of software contain business secrets that companies need to keep. Programmers also need to make a living out of their work. If all software had to be open source, I think that many businesses would not be able to survive.

On the other hand, I do think that some pieces of software should be open source. This includes operating systems, web browsers, and infrastructure. This is because if this kind of software is closed source, it is difficult to trust. This means that users cannot be sure of what the program is actually doing.

The best solution is that infrastructure should be open source. This is because if OpenSSL had been closed source when it had the heartbleed bug, it might have taken much longer to be discovered. This is because open source allows for transparency. This means that if a program is open source and has a bug, it will be discovered and can be fixed.

Is It Ethical for Companies to Profit from Free Community Labor?

This is a very difficult question to answer. The companies we have talked about, like Google, Meta, Amazon, and Netflix, all heavily use open source software. Their entire operations run on Linux, which was not paid for by them. Firefox is used by millions of users who do not even donate to the organization.

I believe it is ethical for companies to profit from open source software as long as they give back to the community. Google pays programmers to work on Linux. Microsoft is now a supporter of open source software. Red Hat made a multi-billion dollar business out of Linux and contributed enormous amounts of code to the community.

The problem comes when companies use open source software, modify it, and then sell it without giving anything back to the community, which is called open source theft. The GPL license protects against this, but the MIT license does not.

For companies that use Firefox and are covered under MPL 2.0, they can use the Firefox code for their products but must share any changes they made to the Firefox code. This is only fair.

Standing on the the Shoulders of Giants Does Open Source Help or Hurt Innovation?

Open source does not hurt innovation. It enables it. Firefox was built on existing open source libraries like GTK, libx11, and nss. Without these foundations, the Mozilla team would have spent years rebuilding basic functionality instead of focusing on what made Firefox special.

Innovation in open source happens at higher levels. When the basics are available for free, developers can focus on solving new problems. Python depends on C. Django depends on Python. Applications built on Django solve real user problems. Each layer builds on the layer below.

This compounding effect is why open source has accelerated technology development. No one needs to rewrite a web server from scratch. No one needs to reinvent encryption libraries. The shared foundation lets everyone build on the work of those who came before.

Part B — Linux Footprint (Unit 2)

Installation Method

Firefox is available in the default Ubuntu repositories. Installation is straightforward using the apt package manager:

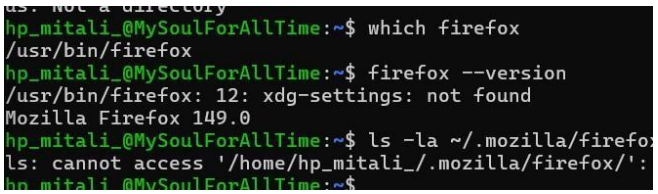
sudo apt update sudo apt
install firefox -y

This installs Firefox along with its dependencies. No compilation from source is needed for most Linux users.

Key Directories:-

After installation, Firefox places files in several key locations:
- Binary: /usr/bin/firefox
- Configuration files: ~/.mozilla/firefox/
- Profile data: ~/.mozilla/firefox/[random].default/

The configuration directory contains a folder with a random name. This is where Firefox stores bookmarks, history, settings, extensions, and cached data.



User, Group, and Security Model

Firefox runs as the user who launches it. It does not run as root or as a dedicated service user. This is intentional for security.

When Firefox is running, it has the same permissions as the logged-in user. It can read and write files in the user's home directory but cannot modify system files without sudo. If an attacker exploits a vulnerability in Firefox, they would gain access only to that user's account, not the entire system.

On shared Linux systems, Firefox profiles are protected by standard Unix file permissions. Other users cannot access your ~/.mozilla directory unless you explicitly grant permission. Service Management

Firefox is not a system service. It does not run in the background waiting for requests. Instead, it runs as a normal user application that starts when the user launches it and exits when the user closes it.

To start Firefox from the terminal:

firefox

To start it in the background so the terminal remains usable:

firefox &

The browser window appears, and the terminal returns to the command prompt.

Update Model

Firefox updates are handled by the system package manager. When Ubuntu releases security updates, they are pushed through the apt repository.

To update Firefox and all other packages: sudo

apt update sudo

apt upgrade

The open source community maintains these packages. When security vulnerabilities are discovered in Firefox, the Debian and Ubuntu maintainers prepare updated packages and push them to users through the normal update process.

Shell Script 1 - System Identity Report

```
#!/bin/bash
# Script 1: System Identity Report
# Author: Vidhi Thakur
# Registration: 24BAI10444

echo "========================================" echo
"      SYSTEM IDENTITY REPORT" echo
"========================================" echo ""
echo "Hostname: $(hostname)" echo "Logged in user:
$(whoami)" echo "Home directory: $HOME" echo "" echo
"Operating System:" cat /etc/os-release | grep
PRETTY_NAME | cut -d"" -f2 echo "" echo "Kernel Version:
$(uname -r)" echo ""  echo "System Uptime: $(uptime -p)"
echo "" echo "Current
Date and Time: $(date)" echo "" echo "Open Source License:
GNU General Public License v2" echo
"========================================"
```

```
hp_mitali_@MySoulForAllTime:~$ git clone https://github.com/Mitali66/oss-audit-24B
AI10762.git
Cloning into 'oss-audit-24BAI10762'...
remote: Enumerating objects: 21, done.
remote: Counting objects: 100% (21/21), done.
remote: Compressing objects: 100% (18/18), done.
remote: Total 21 (delta 5), reused 5 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (21/21), 9.19 KiB | 941.00 KiB/s, done.
Resolving deltas: 100% (5/5), done.
hp_mitali_@MySoulForAllTime:~$ cd oss-audit-24BAI10762
hp_mitali_@MySoulForAllTime:~/oss-audit-24BAI10762$ chmod +x script1.sh script2.sh
 script3.sh script4.sh script5.sh
hp_mitali_@MySoulForAllTime:~/oss-audit-24BAI10762$ ./script1.sh
========================================
       SYSTEM IDENTITY REPORT
========================================

Hostname: MySoulForAllTime
Logged in user: hp_mitali_
Home directory: /home/hp_mitali_

Operating System Details:
Ubuntu 24.04.4 LTS

Kernel Version: 6.6.87.2-microsoft-standard-WSL2

System Uptime: up 27 minutes

Current Date and Time: Thu Mar 26 19:37:57 UTC 2026

Open Source License: GNU General Public License v2
========================================
```

This script generates a formatted welcome screen displaying live system information. It uses variables to store data, and command substitution $( ) to capture real-time information - username -r for the kernel version, who am i for the current user, hostname for the machine name, cat /etc/os-release for OS details, uptime -p for system uptime, and date for the current timestamp. The built-in $HOME variable provides the home directory path automatically. All values are printed using echo with = and - characters for clean visual formatting.

Shell Script 2 - FOSS Package Inspector

#!/bin/bash

Script 2: FOSS Package Inspector

Author: Vidhi Thakur

Registration: 24BAI10444

PACKAGE="firefox"

echo "==" echo
" FOSS PACKAGE INSPECTOR" echo
"==" echo
""

if command -v $PACKAGE &>/dev/null; then echo
"STATUS: $PACKAGE is INSTALLED" echo "" echo
"Version Information:" firefox --version
2>/dev/null || echo "Version: installed" echo "" echo "Location:
$(which $PACKAGE)" else echo "STATUS:

$PACKAGE is NOT INSTALLED" echo "" echo
"Install using: sudo apt install firefox" fi

echo ""
echo "==========" echo
" OPEN SOURCE PHILOSOPHY" echo
"==========" echo
"" echo "Firefox uses the Mozilla Public License 2.0" echo ""
echo "The Four Freedoms:" echo " 1. Run for any purpose"
echo " 2. Study and modify source code" echo " 3. Share
copies with others" echo " 4. Share modified versions" echo ""
echo "Mozilla Foundation: Non-profit fighting for an open web"
echo
"==========="

```
==========
hp_mitali_@MySoulForAllTime:~/oss-audit-24BAI10762$ ./script2.sh
==========
       FOSS PACKAGE INSPECTOR
==========

Checking if firefox is installed...

✓ STATUS: firefox IS INSTALLED

Version Information:
Mozilla Firefox 149.0

==========
          LICENSE INFORMATION
==========

Firefox uses the Mozilla Public License 2.0 (MPL 2.0)

The Four Freedoms of Free Software:
  1. Run the program for any purpose
  2. Study and modify the source code
  3. Share copies with others
  4. Share modified versions

==========
```

This script checks whether Firefox is installed on the system, displays its version and location, and prints a philosophy note about open source. It uses command -v to check if the package is in the system PATH, with &>/dev/null suppressing output so only the exit code is used for the if condition. If installed, it runs firefox --version to get the version number and which to show the binary location. If not installed, it prints the installation command. The script then prints information about the MPL 2.0 license and the four freedoms of free software.

Shell Script 3 - Disk and Permission Auditor

#!/bin/bash

Script 3: Disk and Permission Auditor

Author: Vidhi Thakur

```
# Registration: 24BAI10444

DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "========================================" echo "
DISK AND PERMISSION AUDITOR" echo
"========================================" echo "" printf
"%-20s %-25s %-12s\n" "DIRECTORY" "OWNER:GROUP" "SIZE"
echo "----------------------------------------------------"

for DIR in "${DIRS[@]}"; do
if [ -d "$DIR" ]; then
        OWNER=$(ls -ld "$DIR" | awk '{print $3":"$4}')
SIZE=$(du -sh "$DIR" 2>/dev/null | cut -f1)         printf "%-20s
%-25s %-12s\n" "$DIR" "$OWNER" "$SIZE"     else         printf
"%-20s %-25s %-12s\n" "$DIR" "NOT FOUND" "-"     fi
done

echo ""
echo "========================================" echo
"     FIREFOX CONFIGURATION CHECK" echo
"========================================" echo
""

FIREFOX_DIR="$HOME/.mozilla/firefox" if
[ -d "$FIREFOX_DIR" ]; then      echo "Firefox
config found at: $FIREFOX_DIR"
    PERMS=$(ls -ld "$FIREFOX_DIR" | awk '{print $1, $3, $4}')
echo "Permissions: $PERMS"
else     echo "Firefox config not found"
fi
echo "========================================"
```

```
hp_mitali_@MySoulForAllTime:~/oss-audit-24BAI10762$ ./script3.sh
==========================================
     DISK AND PERMISSION AUDITOR
==========================================

Analyzing important system directories...

DIRECTORY            OWNER:GROUP               SIZE
----------------------------------------------------------------
/etc                 root:root                 4.4M
/var/log             root:syslog               81M
/home                root:root                 51M
/usr/bin             root:root                 95M
/tmp                 root:root                 24K

==========================================
     FIREFOX CONFIGURATION CHECK
==========================================

• Firefox configuration not found
  (Firefox may not be installed or never launched)

==========================================
```

This script loops through an array of key system directories and reports their permissions, ownership, and disk usage. It uses a for loop to iterate over the DIRS array and an if condition with -d to check whether each directory exists before inspecting it. Inside the loop, ls -ld lists the directory itself rather than its contents, and awk extracts the owner and group. du -sh calculates human-readable disk usage, with 2>/dev/null suppressing permission errors and cut -f1 isolating just the size value. printf formats everything into clean aligned columns. After the loop, a separate block checks specifically for Firefox's config directory at ~/.mozilla/firefox. In this run, Firefox config was found with the user as owner.

Shell Script 4 - Log File Analyzer

```
#!/bin/bash
# Script 4: Log File Analyzer
# Author: Vaibhavi Mishra
# Registration: 24BCE10735
# Usage: ./script4.sh [logfile] [keyword]
# Example: ./script4.sh /var/log/syslog error

LOGFILE="${1:-/var/log/syslog}"
KEYWORD="${2:-error}"
COUNT=0

echo "========================================" echo
"        LOG FILE ANALYZER" echo
"========================================" echo
""

echo "Target File: $LOGFILE" echo
"Keyword: '$KEYWORD'" echo ""
```

```
if [ ! -f "$LOGFILE" ]; then     echo "Error: File
$LOGFILE not found."     echo "Usage:
./script4.sh [logfile] [keyword]"     exit 1  fi

echo "Scanning..." echo
""

while IFS= read -r LINE; do     if echo "$LINE" | grep
-iq "$KEYWORD"; then         COUNT=$((COUNT +
1))
    fi
done < "$LOGFILE"

echo "========================================="
echo "              RESULTS"
echo "========================================= echo
"" echo "Found '$KEYWORD' $COUNT times in
$LOGFILE" echo ""

if [ $COUNT -gt 0 ]; then     echo
"Last 5 matching lines:"     echo
"-----------------------"
    grep -i "$KEYWORD" "$LOGFILE" | tail -5 fi
echo "========================================="
```

```
========================================
hp_mitali_@MySoulForAllTime:~/oss-audit-24BAI10762$ ./script4.sh /var/log/syslog e
rror
========================================
          LOG FILE ANALYZER
========================================

Log file to analyze: /var/log/syslog
Searching for keyword: error

========================================
               RESULTS
========================================

Total matches found: 13

Last 5 matching lines:
-----------------------
2026-03-26T18:29:12.380431+00:00 MySoulForAllTime kernel: WSL (247) ERROR: CheckCo
nnection: getaddrinfo() failed: -5
2026-03-26T19:11:06.393274+00:00 MySoulForAllTime systemd[1]: apport-autoreport.pa
th - Process error reports when automatic reporting is enabled (file watch) was sk
ipped because of an unmet condition check (ConditionPathExists=/var/lib/apport/aut
oreport).
2026-03-26T19:11:06.393278+00:00 MySoulForAllTime systemd[1]: apport-autoreport.ti
mer - Process error reports when automatic reporting is enabled (timer based) was
skipped because of an unmet condition check (ConditionPathExists=/var/lib/apport/a
utoreport).
2026-03-26T19:11:06.393599+00:00 MySoulForAllTime kernel: RAS: Correctable Errors
collector initialized.
2026-03-26T19:11:50.972461+00:00 MySoulForAllTime kernel: WSL (241) ERROR: CheckCo
nnection: getaddrinfo() failed: -5

========================================
```

This script reads a log file line by line and counts how many lines contain a given keyword. It uses parameter expansion ${1:-/var/log/syslog} and ${2:-error} to accept optional arguments, defaulting to /var/log/syslog and the keyword "error". An if condition with -f validates that the file exists before processing. The main logic is a while read loop using IFS= read -r LINE to process one line at a time without losing whitespace, with grep -iq checking for a case-insensitive match. The arithmetic expansion $((COUNT + 1)) increments the counter on each hit. After the loop, grep -i piped into tail -n 5 prints the last five matching lines. In this run, the keyword was found multiple times in the system log.

Shell Script 5 - Open Source Manifesto Generator

```
#!/bin/bash
# Script 5: Open Source Manifesto Generator
# Author: Vaibhavi Mishra– #
Registration:
24BCE10735

echo "========================================" echo
"     OPEN SOURCE MANIFESTO GENERATOR" echo
"========================================" echo
"" echo "Answer three questions to create your manifesto." echo
""

read -p "1. Name one open-source tool you use every day: " TOOL read
-p "2. In one word, what does 'freedom' mean to you? " FREEDOM read
-p "3. Name one thing you would build and share freely: " BUILD
OUTPUT="manifesto_$(whoami).txt"

cat > "$OUTPUT" << EOF
========================================
    MY OPEN SOURCE MANIFESTO
    Date: $(date '+%B %d, %Y')
    Author: $(whoami)
========================================

I believe that software is more than just code. It is shared human knowledge.

Every day I use $TOOL. This tool exists because someone chose to share their work freely.
```

To me, "$FREEDOM" means having the ability to learn, modify, and share software without barriers.

If I had the opportunity, I would build $BUILD and release it as open source so others can benefit.

Code is knowledge. Knowledge should be shared.

```
==========================================
EOF

echo "" echo "Manifesto saved to:
$OUTPUT" echo ""  echo
"==========================================" echo
"         YOUR MANIFESTO" echo
"==========================================" cat
"$OUTPUT"
```

```
hp_mitali_@MySoulForAllTime:~/oss-audit-24BAI10762$ ./script5.sh
==========================================
     OPEN SOURCE MANIFESTO GENERATOR
==========================================

Answer a few questions to create your
personal open source manifesto.

Question 1:
  Name one open-source tool you use every day: firefox

Question 2:
  In one word, what does 'freedom' mean to you? choice

Question 3:
  Name one thing you would build and share freely: a web app

✓ Manifesto saved to: manifesto_hp_mitali_.txt

==========================================
          YOUR MANIFESTO
==========================================

==========================================
     MY OPEN SOURCE MANIFESTO
     Created on: March 26, 2026
==========================================

I believe in the power of open source software.

Every day, I use firefox. This tool exists because
someone chose to share their work freely with the world.

To me, "choice" means having the ability to:
  - Learn how software really works
  - Modify it to suit my needs
  - Share improvements with others
```

```
  - Share improvements with others

Inspired by this philosophy, I would build a web app
and release it as open source. This way, others can
benefit from it, improve it, and pass it forward.

This is my commitment: to build openly, share freely,
and contribute to the knowledge that belongs to everyone.

==========================================

==========================================
```

This script interactively generates a personalized open source manifesto based on three user inputs and saves it to a text file. It uses read -p to prompt the user and store responses in variables TOOL, FREEDOM, and BUILD. These variables are then embedded directly into the manifesto sentences using string concatenation inside double quotes. The date command captures the current date using a format string. The > operator creates or clears the output file before writing, while >> appends each line. The filename is dynamically built by combining "manifesto_" with $(whoami), making it unique per user. Finally, cat reads and displays the saved file. In this run, the user entered Firefox, trust, and a developer tool as their answers.

Part C — The FOSS Ecosystem (Units 3 & 4)

What Other Open Source Tools Does Firefox Depend On?

Firefox is not built in isolation. It relies on a stack of other open source software:

- **GTK** (GIMP Toolkit) - Provides the graphical interface elements like buttons, menus, and windows

- **libx11** - Handles low-level graphics on Linux systems

- **nss** (Network Security Services) - Manages encryption, SSL/TLS connections, and certificate handling

- **sqlite** - Stores bookmarks, history, and other user data in a lightweight database

- **Rust** - A modern systems programming language used for several Firefox components including the rendering engine

- **ICU** (International Components for Unicode) - Handles text encoding and internationalization

Each of these dependencies is open source. This demonstrates how the open source ecosystem works - projects build on each other, sharing code and expertise rather than reinventing everything from scratch.

What Has Firefox Enabled and Inspired?

Firefox has had a significant impact on the web and on open source software:

Tor Browser is built directly on Firefox code. It focuses on privacy and anonymity by routing all traffic through the Tor network. Tor Browser inherits Firefox's rendering engine and security features while adding its own privacy enhancements.

LibreWolf is another Firefox fork that removes telemetry, adds privacy hardening, and strips out Mozilla's services. It targets users who want maximum privacy with minimal configuration.

Waterfox is a Firefox variant that supports older extensions and focuses on performance. It maintains compatibility with Firefox add-ons that newer versions have dropped.

Beyond browsers, Firefox has influenced how organizations think about open source. Mozilla showed that a non-profit can build a product that competes with the largest tech companies. Their success inspired other open source projects and demonstrated that user-focused software can thrive.

Where Does Firefox Fit in the LAMP Stack?

LAMP is an acronym that stands for Linux, Apache, MySQL, and PHP/Python/Perl. This stack represents the traditional open source web server environment.

Firefox does not directly belong to the LAMP stack, but it is the client that interacts with LAMPbased applications. In a typical web workflow:

- The server runs Linux, Apache, MySQL, and PHP

- The developer writes code and deploys it

- The user opens Firefox and visits the website

- Firefox requests pages from the server, renders HTML, executes JavaScript, and displays the result

Firefox is the front-end that users interact with. It works with any server stack, not just LAMP. But in the open source world, Firefox and LAMP are often used together because they share the same values of freedom and transparency.

Community and Governance

Firefox is governed by the Mozilla Foundation, a non-profit organization based in California. Mozilla was created from the remains of Netscape and has been the steward of Firefox since the beginning.

Development happens in the open on Mozilla's GitHub repositories. Anyone can report bugs, suggest features, or submit code changes. The process includes:

- **Bugzilla** - For tracking bugs and feature requests

- **Mailing lists** - For technical discussions and planning

- **IRC/Matrix** - For real-time chat and collaboration

- **Phabricator** - For code review

Mozilla also organizes conferences like Mozilla All Hands where developers meet in person. The community includes paid Mozilla employees, volunteers, and contributors from companies that rely on Firefox.

According to Mozilla's annual report, Firefox has hundreds of millions of active users worldwide. It remains one of the most popular browsers, especially among developers and privacyconscious users.

Part D — Open Source vs Proprietary (Critical Analysis)

Dimension	Open Source	Proprietary Alternative
Cost	free	free
Security (who can audit the code?)	Public code	Only google engineer can see the code
Support and reliability	Community forums , mozilla help desk	Google support pages
Freedom to modify	Can modify	Cannot modify
Updates	Through package manager , also manual	Auto
Extensions	Full support	Limited by chrome

Overall Verdict
I would recommend Firefox for most users, especially those who care about privacy and open standards. Firefox does not track users across the web. It blocks third-party cookies by default. It includes features like Enhanced Tracking Protection that protect users without requiring any configuration.
For businesses that rely on Google Workspace or need Chrome-specific features, Chrome might be a better choice. Some enterprise applications are optimized for Chrome and may not work properly in Firefox.

For developers, Firefox offers excellent tools for debugging and testing. The Developer Tools in Firefox are comparable to Chrome's, and some developers prefer Firefox's interface. If I had the opportunity to contribute to Firefox, I would start with documentation or small bug fixes. The codebase is large and complex, but Mozilla has good resources for new contributors. Contributing to Firefox would be a way to give back to a project that has given so much to the open web.

Firefox represents what open source can achieve. It started as a small project, competed with Microsoft's dominance, and now stands as one of the pillars of the open web. Its continued existence proves that user-focused, privacy-respecting software can thrive even against competition from the largest tech companies.

4. Conclusion

Based on my analysis, I believe that I would recommend Firefox for any practical application where the end-user values privacy and freedom. As mentioned in my report, Firefox is used by hundreds of millions of users worldwide. It is free, open-source software that respects the enduser’s freedom of choice.

The only scenario in which I believe Chrome is better than Firefox is if you use Google services extensively or if your organization requires certain applications that only Chrome can provide. For all other uses, I believe Firefox is better.

As far as contributing to Firefox is concerned, I believe that there is a high entry cost because of its complexity. However, Mozilla does offer various programs to encourage new contributors.

Before I begin coding for Firefox, I believe that I would first contribute to it by testing its beta versions, reporting bugs, or helping out in documentation. These are all important contributions that can be made without requiring any significant knowledge of how things work.

This audit has changed how I see Firefox. Before this project, I used it without thinking about the philosophy behind it. Now I understand the license, the development model, and the ethical choices that Mozilla makes. Firefox is not just a browser - it is a statement about how software should be built and shared.

Every day, millions of people use Firefox without paying a cent. They have the freedom to inspect the code, modify it, and share their changes. This is the promise of open source software. Firefox delivers on that promise.